Airbus Pléiades Basic MS ORT: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for Pléiades as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	25/03/2025	Initial draft for inclusion in pathfinder demonstration.
1.0	20/02/2026	Updated draft for pre-publication review. Updates include: • Updated Product Flags & Metadata grade from “Good” to “Excellent” • Minor formatting updates

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1] Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. Available Online.

[RD-2] Airbus Pléiades Imagery User Guide. (Report Number: USRPHR-DT-125-SPOT-2.0, Date of issue: October, 18, 2012)

[RD-3] ESA earth online, missions. See: https://earth.esa.int/eogateway/missions/pleiades

[RD-4] Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016); https://doi.org/10.1038/sdata.2016.18

[RD-5] Vincent Martin, Gwendoline Blanchet, Philippe Kubik, Sophie Lacherade, Christophe Latry, Laurent Lebegue, Florie Lenoir, Florence Porez-Nadal, "PLEIADES-HR 1A&1B image quality commissioning: innovative radiometric calibration methods and results," Proc. SPIE 8866, Earth Observing Systems XVIII, 886610 (23 September 2013); https://doi.org/10.1117/12.2023337

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- Not Assessable means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for Airbus Pléiades is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

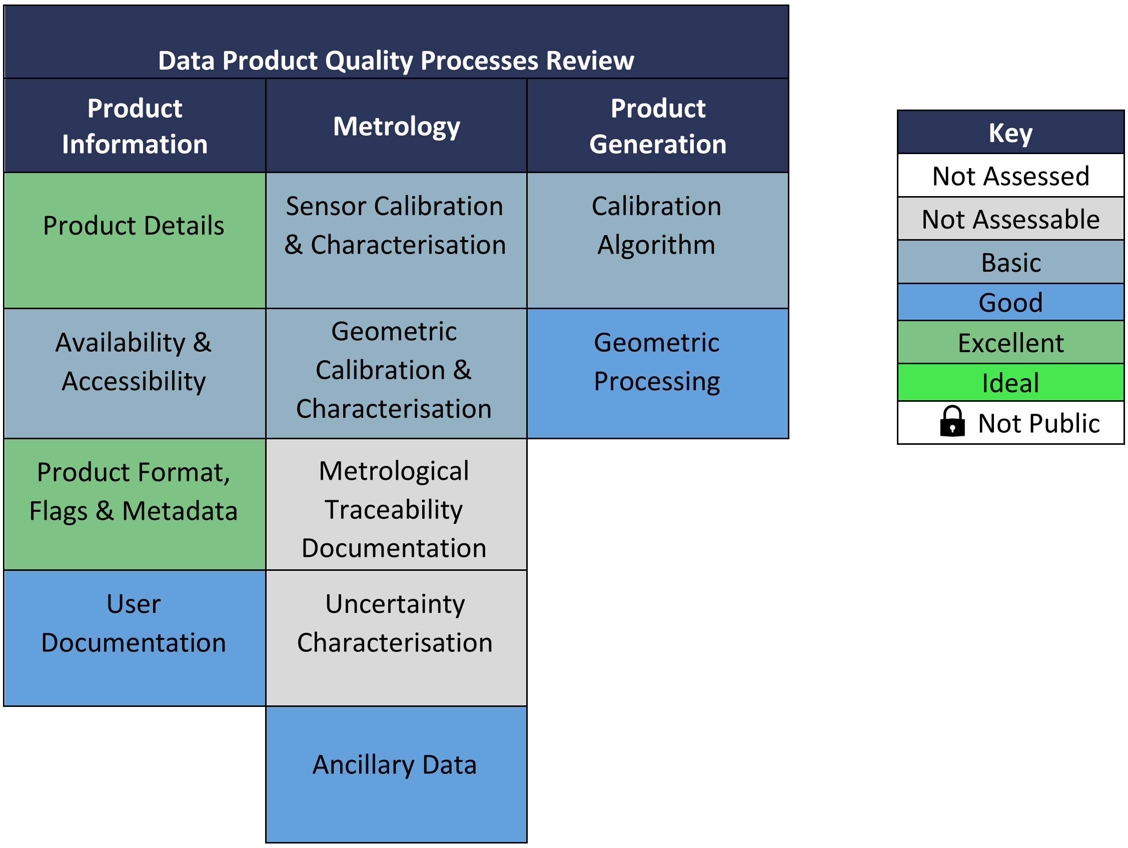

Figure 1 - Data Product Quality Processes Review Maturity Matrix for Airbus Pléiades

3 Product Information

Product Details	
Grade: Excellent	
Justification	*Comprehensive information supplied in the Airbus Pléiades Imagery User Guide and product metadata. Further information, such as Product Locator and Conditions for access and use, are available upon request through Data Supplier's website portal [RD-2].*
Product Name	*Pléiades Basic MS ORT*
Sensor Name	*High Resolution Imager (HiRI) [RD-3]*
Sensor Type	*Optical – Multispectral and Panchromatic (4 bands + PAN)*
Mission Type	*Constellation – 2 satellites*
Mission Orbit	*Sun-synchronous, 10:30 AM descending node, 26-day cycle, 694 km altitude*
Product Version Number	*Included in product metadata, currently: V2.15*
Product ID	*The naming convention uses key information contained in the DIMAP product metadata file. The product directory name is comprised of:* *<DirImage_ID>* *PHR<SAT_NUMBER>_<SPECTRAL_ PROCESSING>_<NUM_PRODUCT>* *Example: PHR1A_PMS_001* *Where:* *<SAT_NUMBER> = {1A, 1B}, Pléiades 1A or Pléiades 1B, or one of them if both used in a mosaic* *<SPECTRAL_PROCESSING> = {P, MS, PMS, MS-N, MS-X, PMS-N, PMS-X}* *<NUM_PRODUCT> = Product index within the Volume. Format is three digits from 001 to 999*
Processing level of product	*Basic (TOA radiance), Ortho (Native pixel values after equalisation, georeferenced image in Earth geometry, corrected from acquisition and terrain off-nadir effects.)*
Measured Quantity Name	*Digital Numbers (DN) / Spectral Radiance*
Measured Quantity Units	*DN / W.st-1.m-2.µm-1*
Stated Measurement Quality	*Radiometric Quality:* *Red / Green / Blue / NIR / PAN < 5 %* *Geometric Quality:* *Location accuracy: 6.5m CE90 @ Nadir*
Spatial Resolution	*Very High Resolution* *Multispectral: 2.8 m GSD (2 m Pixel Size)* *Panchromatic: 0.7 m GSD (0.5 m Pixel Size)*

	Full Swath Width @ Nadir: 20 km
Spatial Coverage	*Global* *Period: 98.79 minutes* *Inclination: 98.2°*
Temporal Resolution	*Revisit capacity: daily*
Temporal Coverage	*Mission lifetime: Minimum of 5 years with an estimated life of more than 10 years* *Launches:* *Pléiades 1A: December 17, 2011;* *Pléiades 1B: December 2, 2012*
Point of Contact	*https://space-solutions.airbus.com/contact-us/*
Product locator (DOI/URL)	*Products available on request via https://space-solutions.airbus.com/contact-us/order-imagery/*
Conditions for access and use	*Depending on the service: https://space-solutions.airbus.com/legal/licences/*
Limitations on public access	*See above licences.*
Product Abstract	*From Airbus Pléiades Imagery User Guide:* *Pléiades is composed of two twin satellites operating as a true constellation on the same orbit and phased 180° from each other. Added to their oblique incidence capability (up to 45° angle) and exceptional agility, this orbit phasing allows the satellites to revisit any point on the globe daily – ideal for anticipating risks, effectively managing crises or for large areas coverage.*

Availability & Accessibility	
Grade: Basic	
Justification	*The product information, such as Product Locator and Conditions for access and use (included in product metadata, etc.) provided meets only some of the FAIR principles.* *Findable, Accessible, Interoperable and Reusable (FAIR) Principles – see [RD-4].*
Compliant with FAIR principles	*No*
Data Management Plan	*Description of data management plan is limited / not available.*
Availability Status	*Products available on request via https://space-solutions.airbus.com/contact-us/order-imagery/*

Product Format, Flags & Metadata	
Grade: Excellent	
Justification	*DIMAP, a documented standard file format, is used. The products also include a good set of documented metadata and data flags but is not CEOS ARD compliant.*
Product File Format	*Products are available in a well-defined file format structure defined in the Appendix of [RD-2]:* *Users can specify the product and image format, from:* • *DIMAP – JPEG 2000 Optimised* • *DIMAP – JPEG 2000 Regular* • *DIMAP – GeoTIFF* • *NITF – JPEG 2000 Regular* • *NITF – GeoTIFF* *DIMAP V2 is the default Pléiades product format. Inside that product, you can select an image format:* • *JPEG 2000: regular compression (8 bits/pixels), recommended for users willing to do some high precision post-processing.* • *JPEG 2000: optimised compression (3.5 bits/pixels), perfect for fast download and easy data sharing.* • *GeoTIFF: uncompressed.*
Metadata Conventions	*DIMAP V2 format*
Analysis Ready Data?	*No*

User Documentation		
Grade: Good		
Justification	There is currently a user guide (see [RD-2]), but there is no ATBD made publicly available.	
Document	*Reference*	*QA4ECV Compliant*
Product User Guide	*Airbus Pléiades Imagery User Guide [RD-2]*	*No*
ATBD	*Documentation not made available to users*	*N/A*

4 Metrology

Sensor Calibration & Characterisation	
Grade: Basic	
Justification	*Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly.* *There is some very high-level information in [RD-2] which outlines the main radiometric corrections/enhancements that have been carried out:* • *Inter-detector equalisation: correction of differences in sensitivity between the detectors (on-board correction).* • *Aberrant detectors correction (none at that time). Panchromatic band restored and de-noised.* • *Pixel sampling at Shannon optimising image quality for downstream value-added processing: Spline kernel resampling into the Primary geometry, zoomed to the factor 7/5 (equivalent resolution of 0.5/2 m in nadir condition).* *Further details on the methods used to evaluate these corrections is not available publicly.*
References	• *RD-2*

Geometric Calibration & Characterisation	
Grade: Basic	
Justification	*Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly.* *There is some very high-level information in [RD-2] which outlines the main geometric processing:* • *The combination of all sub-swaths across in the field of view (20 km nadir condition): synthesis in a virtual focal plane represented by a single linear array for all spectral bands* • *Correction of instrumental and optical distortions: viewing angles adjusted to the single linear array model.* • *Co-registration of all spectral bands: Multispectral and Panchromatic.* • *Attitudes and ephemeris data are refined at ground on the mean estimation:* ○ *Adjustment of the time stamp sampling (along scan line).* ○ *Attitudes filtering over time of acquisition or a posteriori extended over several orbits (aka Refined Attitude Data).* • *Consistent alignment of the physical model ancillary data and RPC analytic model data.*

	Further details on the methods used to evaluate these corrections is not available publicly.
References	• *RD-2*

Metrological Traceability Documentation	
Grade: Not Assessable	
Justification	*There are no metrology documentation or traceability chain / uncertainty tree diagrams publicly available.*
References	

Uncertainty Characterisation	
Grade: Not Assessable	
Justification	*Information is given in Appendix B of the user guide [RD-2] on the performance regarding image quality, with reference to having used pre-flight calibration in the characterisation.* *However, there is no detailed information regarding the methods used to characterise the uncertainties publicly available.*
References	• *RD-2 (Appendix B)*

Ancillary Data	
Grade: Good	
Justification	*An overview of ancillary data for the whole collection is available in the user guide [RD-2] and per product ancillary data is available (e.g. viewing and solar angles, longitude, latitude) in product metadata. However, there is no detailed information regarding uncertainty characterisation for any of the ancillary data.* *Specifically, DEM data used in ortho-rectification is specified at collection level, and not per product. In the user guide [RD-2], there is an overview of the geometric details used in the ORTHO product:* • *Geographic projections: WGS84 – latitude/longitude sampling according to ARC or DTED2 zoning standards (please refer to A.5.1 for more details)* • *Mapping projection: Most of the projections registered by EPSG, in metres (please refer to A.5.2 for more details)* • *GCP: Reference3D Ortho layer* • *DEM: Reference3D DEM layer (DTED2), SRTM (DTED1), GLOBE (DTED0)*
References	• *RD-2*

5 Product Generation

Calibration Algorithm	
Grade: Basic	
Justification	*Information on the radiometric and atmospheric corrections given in Appendix D.4 of the User Guide with description of input parameters and where to access them in the XML metadata file. However, not thorough enough to deem "fit for purpose" in terms of the mission's stated performance.*
References	• *RD-5*

Geometric Processing	
Grade: Good	
Justification	*Information on the geometric processing given in Appendix C of the User Guide with description of input parameters and where to access them in the XML metadata file.* *The described algorithm is reasonable for the mission's stated performance and the calibration quality is considered sufficient.*
References	• *RD-2*